

Generatiive AI in life steward — AIVA



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Group Number: 62 Module Name: CPT208 Human-Centric Computing

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Background and Introduction

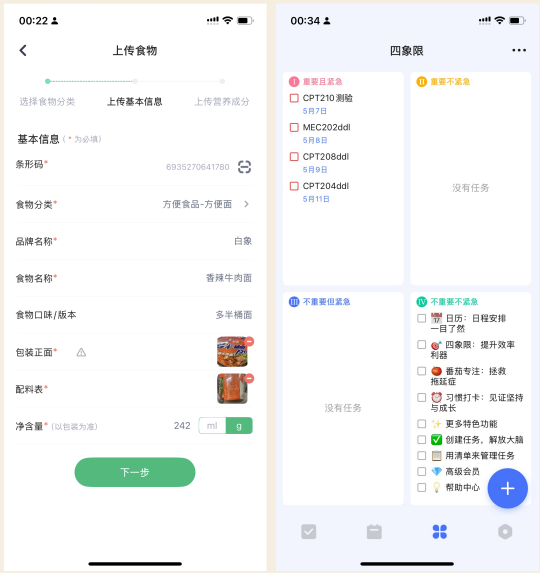
Integrated solutions to avoid fragmentation

Modern life management faces key challenges: traditional tools (such as ordinary calendar apps and static diet plans) cannot adapt to **dynamic personal needs**, **real-time scenarios** and overall health goals.

System Type	Representative Tools	Features	Existing Problems
Schedule Management	Google Calendar, Notion	Powerful recording function, weak reasoning ability	Unable to actively recommend or integrate factors such as healthy diet
Diet and Nutrition Decision - Making Assistance	MyFitnessPal, Yazio	Detailed data but cumbersome input	Lacks visual recognition and contextual suggestions
	ChatGPT (Dialogue), Mental Health	Can generate answers	Lacks integration of real - life data and continuous tracking

Human-centered computing: Integration and Personalization

People increasingly want personalized help for health, efficiency, and decision - making. Yet current solutions are **fragmented** and **context - unaware**. AIVA, an AI life assistant, simplifies daily tasks via adaptive **nutrition tracking, smart scheduling, proactive decision - making and so on**. It merges **real - time** data analysis with human - centric design, swapping rigid tools for personalized support to foster habits, optimize tasks, and encourage healthier, more intentional choices.

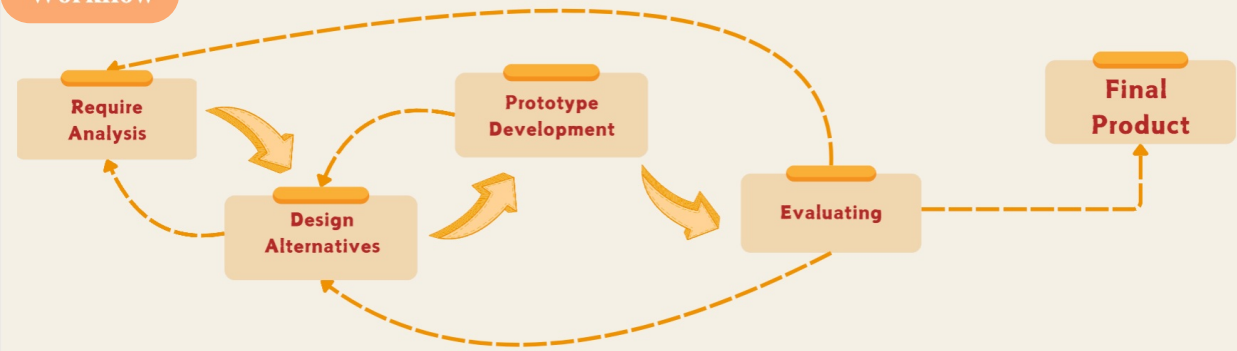


Example

Design and Methodology

Target Users College students and middle-aged groups lacking self-discipline

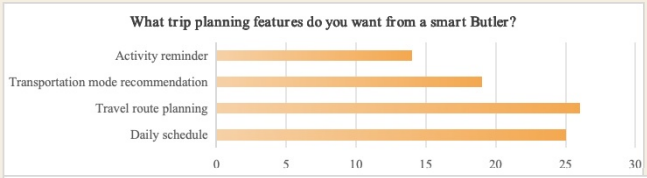
Workflow

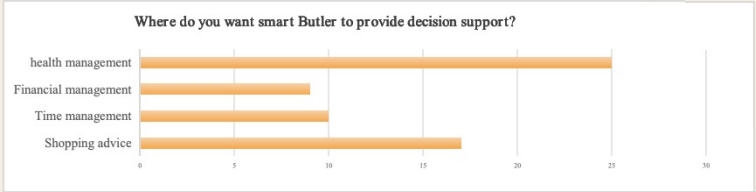


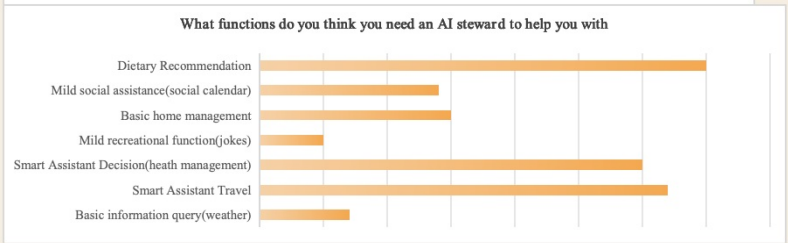
Requirement analysis serves as the starting point. By conducting research, it clarifies user needs and pain points, thus anchoring the direction for product development. **Designing alternatives, prototype development**, and **evaluation form** a coherent chain. Exploring multiple solutions, building prototypes, and conducting comprehensive evaluations help identify and solve problems continuously. The results of **evaluation are fed back to the previous stages** for **cyclic optimization**. This enables the product to be refined step by step, resulting in a **final product** that meets the needs of the market and users.

Require Analysis

An online survey with both open-end and closed-end questions and received 70 valid results.









In trip planning, the bar chart shows high demand for **daily schedule** and **travel route planning**, indicating users want smart Butlers to help organize travel efficiently.

The bar chart on decision support reveals a strong need for **health management**, highlighting users' eagerness for professional health - related advice.

The word cloud emphasizes terms like "DAILY SCHEDULE", "FUNCTIONAL DIVERSITY", and "ACCURACY OF ALGORITHM", suggesting users expect **smart Butlers with diverse, accurate-performing features** for good service.

Require Analysis

An online survey with both open-end and closed-end questions and received 70 valid results.



Alex **"The Health Seeker"**

Digital Nomad

Alex is a 30-year-old freelance designer who works remotely from different cities. He has basic cooking skills but struggles to maintain healthy eating habits while traveling. His diet consists mostly of cafe meals and grocery store snacks. Alex wants to improve his nutrition without sacrificing his flexible lifestyle.

Goals

- Eat balanced meals despite irregular schedules
- Discover local healthy food options when traveling
- Learn quick meal prep techniques for small kitchens
- Reduce food waste while moving frequently

Where AI Steward Can Help

Dietary Recommendation
- Suggest portable kitchen tools for travelers
- Recommend meal kits based on current location
- Adjust recipes for minimal equipment needs

Schedule Integration
- Sync with travel itinerary to plan meal times
- Locate grocery stores near accommodations
- Remind to use perishable ingredients first

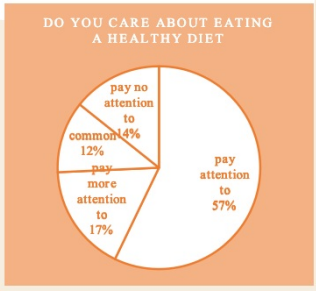
Decision Support
- Compare nutrition values of local cuisine options
- Calculate cost/benefit of cooking vs eating out
- Suggest optimal meal timing across time zones

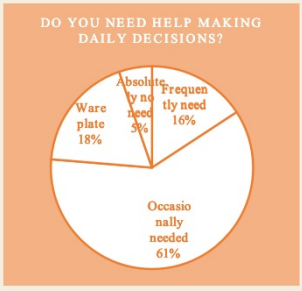
Progress Tracking
- Map nutrition intake across locations
- Celebrate streak of healthy choices
- Adjust recommendations based on energy levels

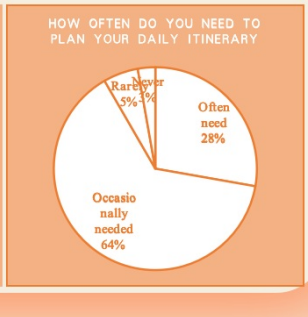
As a digital nomad, Alex can't maintain **healthy eating** while traveling. His need for **better nutrition without sacrificing flexibility** shows the demand for Aiva app's **personalized diets**. Alex wants balanced meals and local healthy food. Aiva app's **dietary recs and schedule integration** can meet his healthy - eating needs in various situations.

Facing challenges, Alex can benefit from Aiva app's **decision - support and progress - tracking** to make healthy choices, proving the app's importance.

The pie charts show that over half care about healthy diets, most need occasional help with **daily decisions**, and the majority either often or occasionally plan their **daily itineraries**, indicating varied yet significant demands for **health - related attention** and **organizational assistance** in daily life.

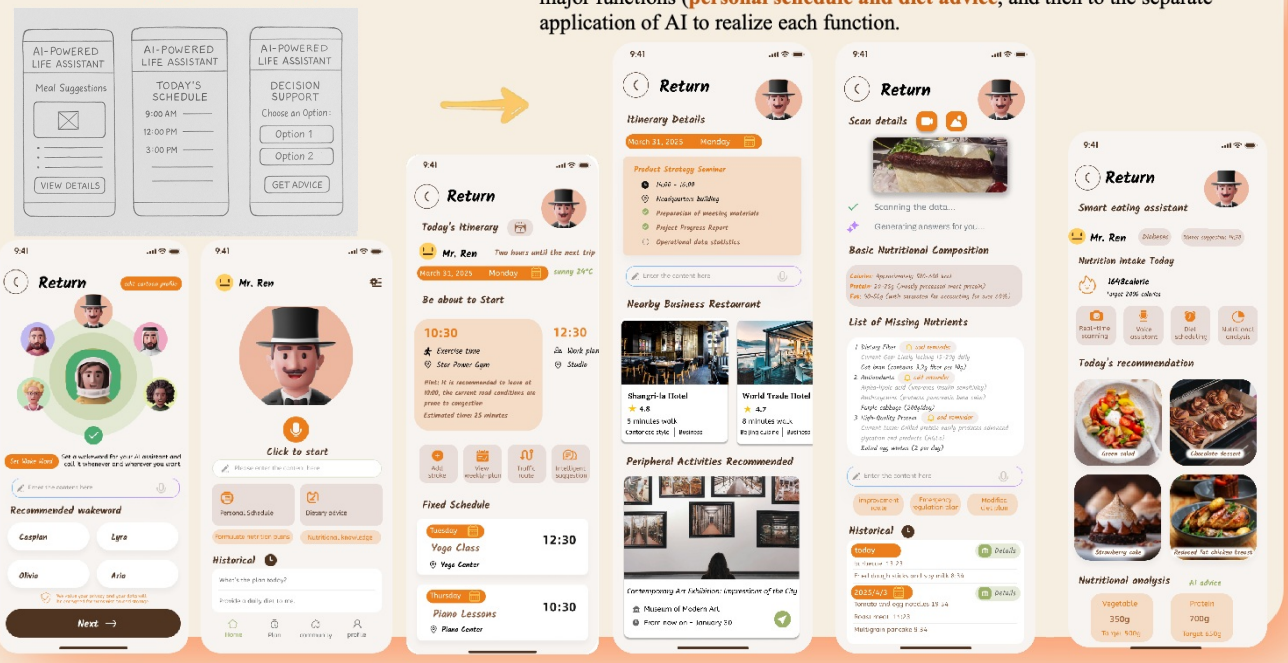






Prototype Implement

From the **setting of cartoon characters and wake words**, to the **intelligent decision-making function** on the main page and the entry points of the other two major functions (**personal schedule and diet advice**, and then to the separate application of AI to realize each function.



Iteration 1

Require Analysis

① The **lack** of **login and pop-up pages** makes the **image of the butler** not deeply rooted in people's minds

② Some buttons with **extra** functions are **duplicated** with the previous page functions



Prototype Implement



When users **browse** the heart-fluttering routes on the **Ctrip Travel app**, AIVA will introduce **relevant information** such as the scenery along the way to users in the form of **pop-up Windows** with the **users' permission**. On the jump page, **an image of a butler** is given, allowing users to better **experience the feeling of a butler**. The **default one-click login** on the mobile phone is convenient for users to operate

Iteration 2

Require Analysis

① The current settings of cartoon characters **cannot well reflect AI-generated images, and their self-creativity is not that strong**.

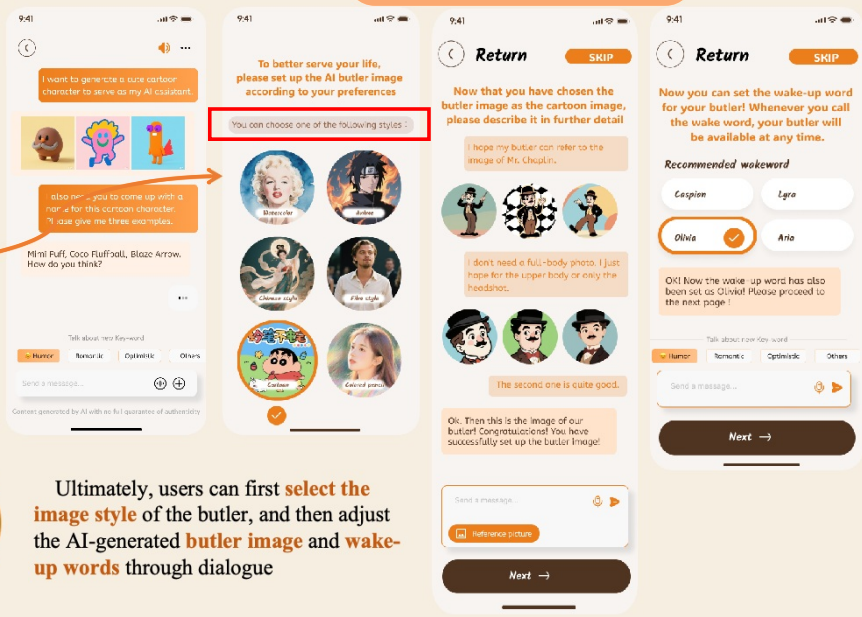
The pages featuring cartoon characters and wake words are rather **mixed together**, not very intuitive and **aesthetically pleasing**.

② The sense of **companionship** of the butler needs to be **enhanced**.

The generated **butler image** will **frequently appear** on subsequent pages and can also accompany users in the form of **pop-up Windows** or **floating Windows**



Prototype Implement



Ultimately, users can first **select the image style** of the butler, and then adjust the AI-generated **butler image** and **wake-up words** through dialogue

Iteration 3

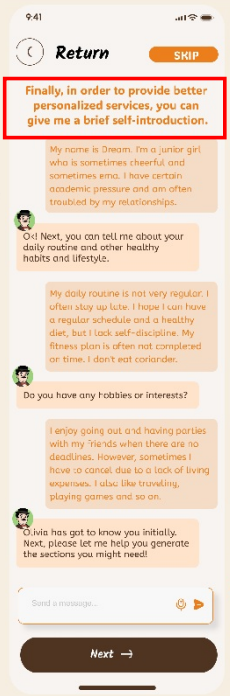
Require Analysis

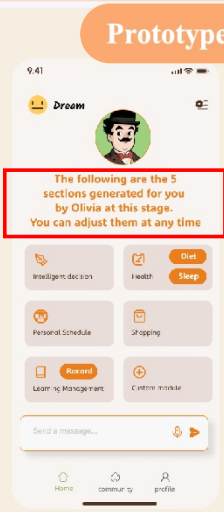
① Diet, schedule and decision-making are just preset **sections that are relatively fixed**. As a life manager, one **cannot comprehensively manage the user's life**.

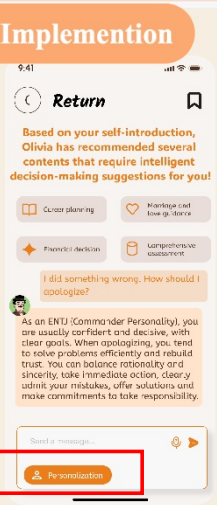
② The current intelligent decision-making is **not personalized** enough and fails to reflect users' behavioral logic, thinking patterns, etc.

③ On pages **lacking a personal growth page**, users cannot get an **overview of their recent lives**.

Prototype Implemention









AIVA can learn more about **user information** through **guided Q&A**, such as MBTI, living habits, daily activities, etc. Based on the **analysis of users**, it **generates the sections** that users need. Users can also **customize the sections**.

On the decision-making page, users can also **make personalized decisions** by **choosing the decision-making mode** and behavioral logic that consider their personal habits.

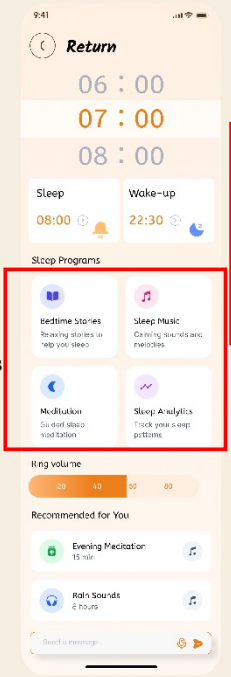
Iteration 4

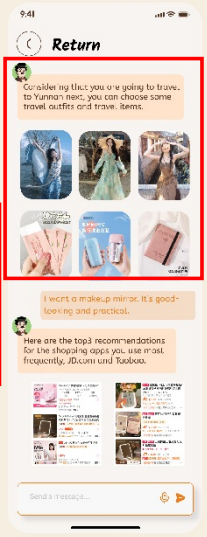
Require Analysis

① For the majority of AIVA users, the **sleep and shopping sections** are also very important. Therefore, users hope that AIVA can also present these pages as **preset sections with specific pages**.

② On the diet page, after giving suggestions based on the analysis of nutritional components, the suggestions **only provide the amount of nutrients** but no specific food recommendations. Users **cannot know exactly what to eat**.



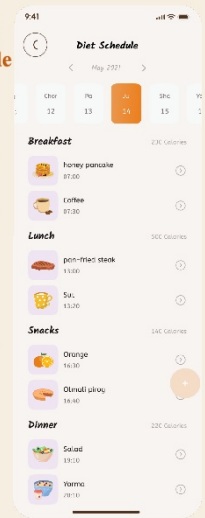




Prototype Implemention

In sleep management, AIVA can **analyze users' sleep** and **provide suitable lullabies** for them. In the shopping section, AIVA can **recommend good products** based on users' schedules and make **intelligent comparisons**.

Based on the analysis of nutritional components and the **dietary plan recommendations** on how to consume specific nutrients, users can **obtain healthy diet advice more intuitively**.



Iteration 5

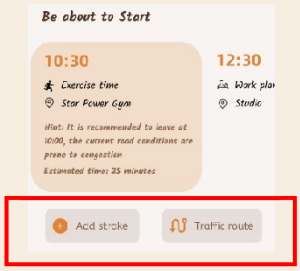
Require Analysis

① The icon of the app is **not warm enough** and has a **robot-like** feel.





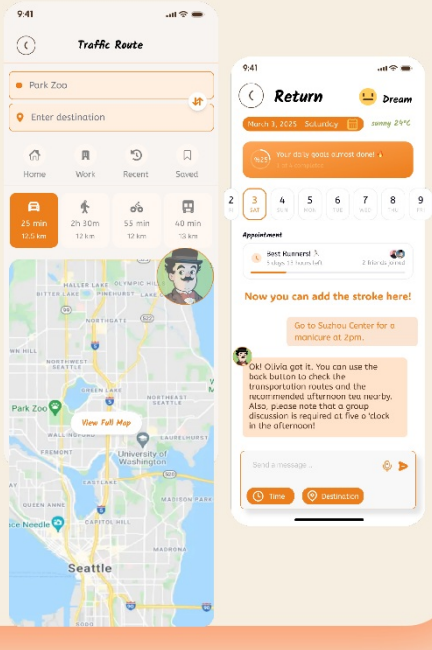
② The specific pages of **add stroke and traffic route are lacking**



The new APP avatar can better reflect **a sense of a butler**, with a more **intimate and warm** feeling

AIVA can **proactively provide transportation routes** when users have relevant itineraries. Users can also **simply and conveniently modify their itineraries** through interaction with the butler.

Prototype Implemention



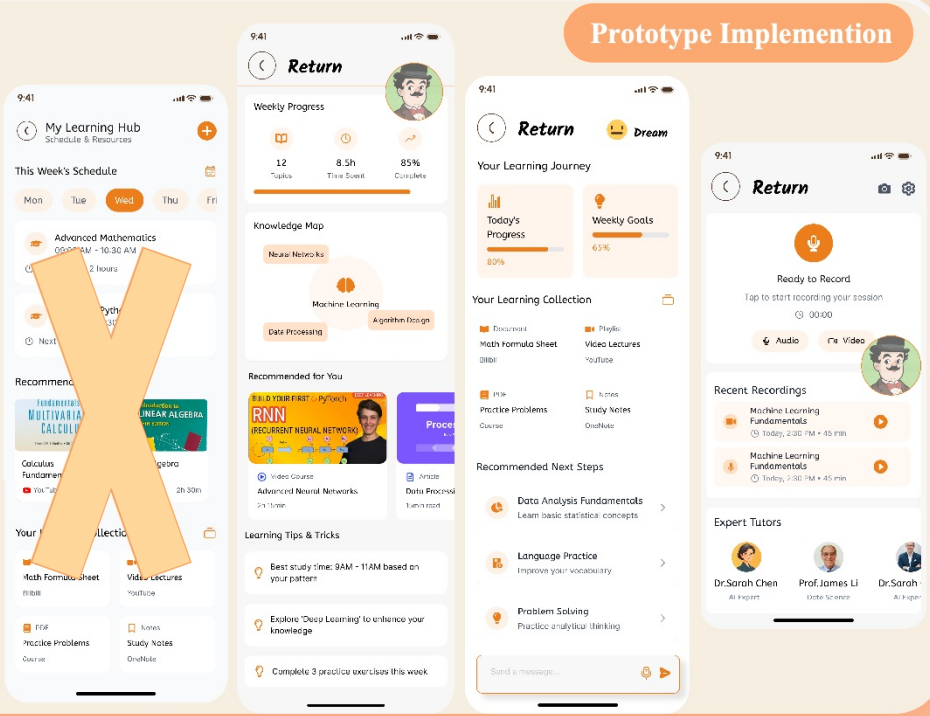
Iteration 6

Require Analysis

As one of the main user groups of AIVA, the student group also has a very important aspect of **learning**. Users have reported that another function of the camera can be utilized: **intelligent recording of class time, recording key points** of the class, and after class, **plans can be made and relevant textbooks recommended** to improve the academic performance of users as students.

If the class schedule is written here again, it will **conflict with the schedule arrangement**, so this page is omitted.

Prototype Implemention



Evaluation and Results

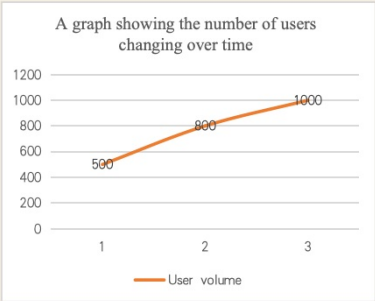
Evaluation method

Heuristic evaluation: Based on the principle of usability, self-check whether the application interface and functions are easy to use.
Usability test: Recruit 100 users from different backgrounds, including students, professionals, etc.
Data analysis: Integrate tools in the application to collect indicators such as the number of daily active users, the frequency of function usage, and user retention rate.

Evaluation result

Quantitative result **User activity:** The average daily active users were 500 in the first month of launch, increased to 800 in the second month, and reached 1,000 in March.

Function usage: The diet decision function is used most frequently, accounting for 35% of the function visits. The schedule function accounts for 25%. Sleep management and learning functions account for 20% and 15% respectively.



To **compare the efficiency of users** in completing tasks before and after the iteration of AIVA, three typical tasks, namely formulating a diet plan, arranging a weekly schedule and purchasing goods, were selected for testing. Draw a line graph with the **iterative version as the abscissa** and the average time to complete the task as the ordinate. Data shows that **before the iteration**, the average time for users to complete the formulation of a diet plan was **15 minutes**. **After three iterations**, it was shortened to **8 minutes**. The average time for arranging a weekly schedule has decreased from **12 minutes to 6 minutes**. The average time for purchasing goods has decreased from **20 minutes to 10 minutes**, clearly demonstrating the **significant improvement in users' task completion efficiency** after the iteration of AIVA.

Evaluation and Results

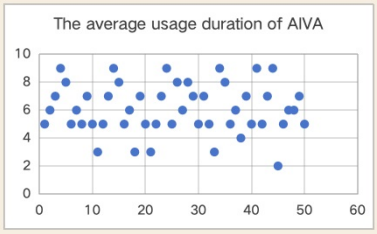
Evaluation result

Qualitative feedback **Positive**: Users love the **personalized features** of applications, especially the ability to **customize AI butlers**. The functional integration is **convenient**. Managing multiple aspects of life in one application can **save time**.

Negative: Some users feel confused when switching between different functional sections and are also concerned about **data privacy**, hoping for more transparent data usage.







Humanistic influence

Improving health management level: Take Alex as an example. After using AIVA for three months, his weight dropped from 75 kilograms to 70 kilograms and his body fat percentage decreased from 25% to 22%.
Improving life efficiency: A survey of 50 users found that they saved an average of 5 to 8 hours per week after using AIVA.
Social impact: AIVA is expected to drive the **development of AI life management apps**.

Discussion and Reflections

Project outcomes and Objectives

Advantages: Personalized customization (customizing AI butlers, exclusive services, making decisions based on user behavior patterns) Diverse functions (covering all aspects of life)
Deficiency: Reliance on data; The functions are **complex and difficult** to master
Compare with the existing schemes: It is **highly personalized and integrated**, outperforming most single-function applications. However, mature applications may have an advantage in terms of the number of users and interface

Team and individual reflection

Team **Advantages: Smooth communication** and **complementary expertise** among members
Improvement: Strengthen early **user testing** and optimize resource allocation

Individual **Gain: Knowledge of AI integration**; The design process of **figma** User-centered design concept (**I was mainly responsible for the functional realization of figma, page production and the overall framework design of the poster**)
Improvement: Enhance the ability to **coordinate resources** in complex projects and be **more proactive** in participating in the early testing planning

Ethics and Principles

Protect data; Privacy Avoid AI bias

Acknowledgment

- I'm deeply grateful to my **CPT208 instructors, Li Yue and Ma Teng**, for their guidance on **human - centric design principles.** Their feedback on ethical AI implementation during the project evaluation was invaluable.
- I also thank my **TA, Shuhao Zhang**, for his support.
- Thanks to the **testers** for their participation in line with participatory design.
- Appreciation to my **team members** for our early - stage collaboration.
- Finally, kudos to the developers of **open - source tools (TensorFlow, ARCore)** for facilitating AIVA's technical realization.

Reference

[1] CPT208 Lecture Notes, "Human-Centered Design Methods," University of Xi'an Jiaotong-Liverpool, 2025.
[2] Group 62, "Generative AI in Life Steward – AIVA Project Poster," CPT208, 2025.